Lab 04
Internet Security
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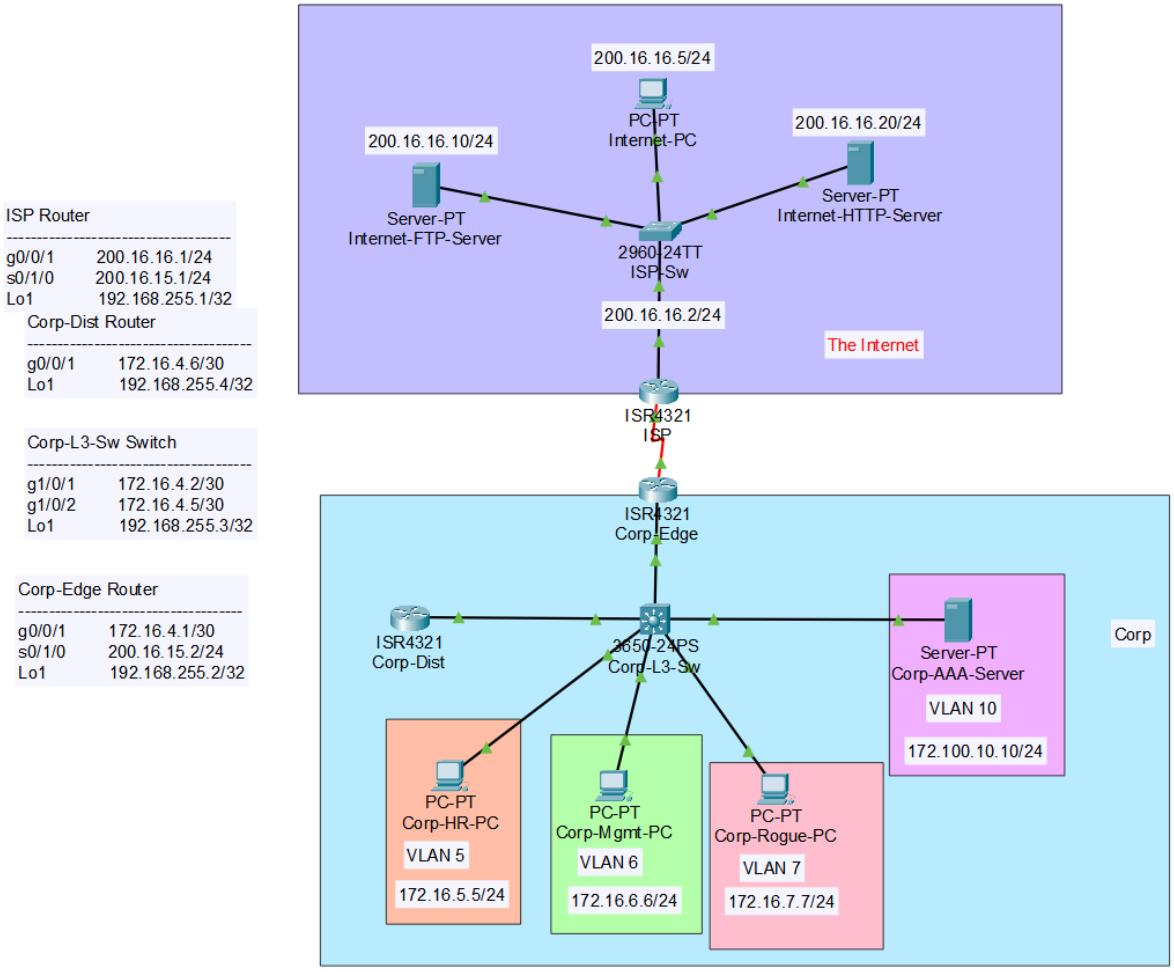
Description:

This lab focuses on the implementation of zone based firewall technology, access control lists, and NAT/PAT to try to protect networks from external threats coming from the internet. Limiting what protocols can be used, mitigating spoofing attacks, and how packet filters can be used to mitigate attacks are all explored in this lab.

Topology:



Key Syntax:

Command	Function
enable	Enters privileged EXEC mode
configure terminal	Enters global configuration mode
hostname [name]	Sets the hostname of the device
enable secret [password]	Sets an encrypted privileged mode password
interface [interface]	Enters interface configuration mode
ip address [ip_address] [subnet_mask]	Configures the IP address and subnet mask on an interface
clock rate [rate]	Sets the clock rate on a serial interface
no shutdown	Enables the specified interface
username [username] secret [password]	Creates a username with an encrypted password
service password-encryption	Encrypts plaintext passwords in the configuration
banner motd [message]	Sets a message of the day banner
router ospf [process_id]	Enters OSPF configuration mode
network [network] [wildcard_mask] area [area_id]	Configures OSPF routing for a network
exit	Exits the current configuration mode
end	Exits configuration mode and returns to privileged EXEC mode
vlan [vlan_id]	Creates a VLAN
name [name]	Assigns a name to a VLAN
interface vlan [vlan_id]	Enters VLAN interface configuration mode
spanning-tree mode [mode]	Configures the spanning-tree mode
switchport mode [mode]	Configures the interface mode (access/trunk)
switchport access vlan [vlan_id]	Configures the VLAN membership for an access port
zone security [name]	Creates a security zone
zone-pair security [name] source [source_zone] destination [destination_zone]	zone-pair security [name] source [source_zone] destination [destination_zone]

class-map type inspect match-any [name]	Creates a class-map to match traffic based on criteria
match protocol [protocol]	Specifies the protocol to match in a class-map
policy-map type inspect [name]	Creates a policy-map to define actions for matched traffic
class type inspect [name]	Enters class-map configuration mode within a policy-map
inspect	Enables inspection of matched traffic in a policy-map
zone-member security [zone]	Assigns an interface to a security zone
show policy-map type inspect zone-pair sessions	Displays firewall sessions for a specific zone pair
ping <ip>	Sends an ICMP echo request to the specified IP address.
access-list <number> permit/deny <network> <wildcard_mask>	Configures an access control list (ACL).
ip access-group <number> in/out	Applies an ACL to an interface.
router ospf <process_id>	Enables OSPF routing and enters router configuration mode.
network <network> <wildcard_mask> area <area_id>	Defines which networks are advertised in OSPF.
line vty 0 4	Enters line configuration mode for the virtual terminal lines (used for remote login).
password <password>	Sets the password for the line.
login	Requires a password for login.

Verification:

Here is the OSPF that I have configured On the Rotuers and Switches

Corp-Dist

Physical Config CLI Attributes

IOS Command Line Interface

```
                ^
% Invalid input detected at '^' marker.

Corp-Dist#sh ip ospf
 Routing Process "ospf 1" with ID 192.168.255.4
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 1. Checksum Sum 0x000363
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 2
        Area has message digest authentication
        SPF algorithm executed 6 times
        Area ranges are
        Number of LSA 5. Checksum Sum 0x023aeb
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

Corp-Dist#
```

Corp-Edge

Physical Config CLI Attributes

IOS Command Line Interface

```
Corp-Edge#sh router ospf
                ^
% Invalid input detected at '^' marker.

Corp-Edge#sh ip ospf
 Routing Process "ospf 1" with ID 192.168.255.2
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 1. Checksum Sum 0x000363
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 3
        Area has message digest authentication
        SPF algorithm executed 5 times
        Area ranges are
        Number of LSA 5. Checksum Sum 0x023aeb
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

```
Corp-L3-Sw>enable
Password:
Corp-L3-Sw#sh ip ospf
 Routing Process "ospf 1" with ID 192.168.255.3
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 1. Checksum Sum 0x000363
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 7
        Area has message digest authentication
        SPF algorithm executed 5 times
        Area ranges are
        Number of LSA 5. Checksum Sum 0x023aeb
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

Corp-L3-Sw#
```

```
Corp-L3-Sw#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.4.1 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       172.16.4.0/30 is directly connected, GigabitEthernet1/0/1
C       172.16.4.4/30 is directly connected, GigabitEthernet1/0/2
C       172.16.5.0/24 is directly connected, Vlan5
C       172.16.6.0/24 is directly connected, Vlan6
C       172.16.7.0/24 is directly connected, Vlan7
     172.100.0.0/24 is subnetted, 1 subnets
C       172.100.10.0 is directly connected, Vlan10
     192.168.255.0/32 is subnetted, 3 subnets
O       192.168.255.2 [110/2] via 172.16.4.1, 00:49:54, GigabitEthernet1/0/1
C       192.168.255.3 is directly connected, Loopback1
O       192.168.255.4 [110/2] via 172.16.4.6, 00:50:04, GigabitEthernet1/0/2
O    200.16.15.0/24 [110/65] via 172.16.4.1, 00:49:54, GigabitEthernet1/0/1
O*E2 0.0.0.0/0 [110/1] via 172.16.4.1, 00:49:54, GigabitEthernet1/0/1

Corp-L3-Sw#
```

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```
Corp-Dist#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.4.5 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 6 subnets, 3 masks
O       172.16.4.0/30 [110/2] via 172.16.4.5, 00:50:12, GigabitEthernet0/0/1
C       172.16.4.4/30 is directly connected, GigabitEthernet0/0/1
L       172.16.4.6/32 is directly connected, GigabitEthernet0/0/1
O       172.16.5.0/24 [110/2] via 172.16.4.5, 00:50:22, GigabitEthernet0/0/1
O       172.16.6.0/24 [110/2] via 172.16.4.5, 00:50:22, GigabitEthernet0/0/1
O       172.16.7.0/24 [110/2] via 172.16.4.5, 00:50:22, GigabitEthernet0/0/1
     172.100.0.0/24 is subnetted, 1 subnets
O       172.100.10.0/24 [110/2] via 172.16.4.5, 00:50:22, GigabitEthernet0/0/1
     192.168.255.0/32 is subnetted, 3 subnets
O       192.168.255.2/32 [110/3] via 172.16.4.5, 00:50:12, GigabitEthernet0/0/1
O       192.168.255.3/32 [110/2] via 172.16.4.5, 00:50:22, GigabitEthernet0/0/1
C       192.168.255.4/32 is directly connected, Loopback1
O    200.16.15.0/24 [110/66] via 172.16.4.5, 00:50:12, GigabitEthernet0/0/1
O*E2 0.0.0.0/0 [110/1] via 172.16.4.5, 00:50:12, GigabitEthernet0/0/1

Corp-Dist#
```

```
ISP>enable
Password:
ISP#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 200.16.15.2 to network 0.0.0.0

     192.168.255.0/32 is subnetted, 1 subnets
C       192.168.255.1/32 is directly connected, Loopback1
     200.16.15.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.16.15.0/24 is directly connected, Serial0/1/0
L       200.16.15.1/32 is directly connected, Serial0/1/0
     200.16.16.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.16.16.0/24 is directly connected, GigabitEthernet0/0/1
L       200.16.16.1/32 is directly connected, GigabitEthernet0/0/1
S*   0.0.0.0/0 [1/0] via 200.16.15.2

ISP#
```

After configuring static routes and OSPF test for connectivity

Fire	Last Status	Source	Destination	Type
	Successful	Internet-PC	Corp-Edge	ICMP
	Successful	Internet-PC	Corp-Dist	ICMP
	Successful	Internet-PC	Corp-Mgmt-PC	ICMP
	Successful	Internet-PC	Corp-AAA-Server	ICMP
	Successful	Corp-Dist	Corp-Rogue-PC	ICMP
	Successful	Corp-HR-PC	Corp-Rogue-PC	ICMP
	Successful	Corp-HR-PC	Internet-FTP-Server	ICMP
	Successful	Corp-Rogue-PC	Internet-HTTP-Server	ICMP
	Successful	Internet-FTP-Server	Corp-Edge	ICMP
	Successful	Internet-FTP-Server	Corp-Mgmt-PC	ICMP
	Successful	Corp-HR-PC	Internet-PC	ICMP

Pings after Nat and Pat, the corp

Fire	Last Status	Source	Destination	Type	Color	Time(sec	Periodic	Num	Edit
	Successful	Corp-HR-PC	Corp-Edge	ICMP		0.000	N	0	(edit)
	Successful	Corp-HR-PC	Corp-Mgmt-PC	ICMP		0.000	N	1	(edit)
	Successful	Corp-HR-PC	Internet-PC	ICMP		0.000	N	2	(edit)
	Successful	Internet-PC	Corp-HR-PC	ICMP		0.000	N	3	(edit)
	Successful	Corp-HR-PC	Corp-Dist	ICMP		0.000	N	4	(edit)
	Successful	Corp-HR-PC	Corp-Rogue-PC	ICMP		0.000	N	5	(edit)
	Successful	Internet-FTP-Server	Corp-Rogue-PC	ICMP		0.000	N	6	(edit)
	Successful	Corp-Rogue-PC	Internet-FTP-Server	ICMP		0.000	N	7	(edit)
	Successful	Corp-Mgmt-PC	Internet-PC	ICMP		0.000	N	8	(edit)
	Successful	Internet-FTP-Server	Corp-Rogue-PC	ICMP		0.000	N	9	(edit)
	Successful	Corp-Rogue-PC	Internet-HTTP-Server	ICMP		0.000	N	10	(edit)
	Successful	Corp-HR-PC	Internet-PC	ICMP		0.000	N	11	(edit)

Here are the nat translations tables for the Corp-Dist router and the Corp-edge router

Corp-Dist

```
UNAUTHORIZED ACCESS PROHIBITED!!!

User Access Verification

Password:

Corp-Dist>enable
Password:
Corp-Dist#sh ip nat
% Incomplete command.
Corp-Dist#sh ip nat translations
Corp-Dist#sh ip nat translation
Corp-Dist#
Corp-Dist#sh ip nat translations
Pro  Inside global      Inside local        Outside local       Outside global
icmp 200.16.16.100:20   172.16.4.6:20       172.16.5.5:20       172.16.5.5:20
icmp 200.16.16.100:2    172.16.4.6:2        172.100.10.10:2     172.100.10.10:2
icmp 200.16.16.100:2    172.16.4.6:2        200.16.15.1:2       200.16.15.1:2
icmp 200.16.16.100:5    172.16.4.6:5        200.16.16.10:5      200.16.16.10:5
```

Corp-Edge

IOS Command Line Interface

```
Corp-Edge#sh ip nat translations
Corp-Edge#sh ip nat translations
Corp-Edge#sh ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 200.16.16.100:1   172.16.4.1:1       172.16.4.2:1       172.16.4.2:1
icmp 200.16.16.100:2   172.16.4.1:2       172.16.6.6:2       172.16.6.6:2
icmp 200.16.16.100:3   172.16.4.1:3       172.16.7.7:3       172.16.7.7:3

Corp-Edge#sh ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 200.16.16.100:1   172.16.4.1:1       172.16.4.2:1       172.16.4.2:1
icmp 200.16.16.100:2   172.16.4.1:2       172.16.6.6:2       172.16.6.6:2
icmp 200.16.16.100:3   172.16.4.1:3       172.16.7.7:3       172.16.7.7:3

Corp-Edge#sh ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 200.16.16.100:1   172.16.4.1:1       172.16.4.2:1       172.16.4.2:1
icmp 200.16.16.100:2   172.16.4.1:2       172.16.6.6:2       172.16.6.6:2
icmp 200.16.16.100:3   172.16.4.1:3       172.100.10.10:3    172.100.10.10:3
icmp 200.16.16.100:3   172.16.4.1:3       172.16.7.7:3       172.16.7.7:3

Corp-Edge#
```

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ISP router ACLs

ISP

```
Password:

ISP>enable
Password:
ISP#sh ip access-lists
Standard IP access list DENY_INTERNAL
    10 deny 10.0.0.0 0.255.255.255
    20 deny 172.16.0.0 0.15.255.255
    30 deny 192.168.0.0 0.0.255.255
    40 deny 127.0.0.0 0.255.255.255
    50 permit any

ISP#sh access-lists
Standard IP access list DENY_INTERNAL
    10 deny 10.0.0.0 0.255.255.255
    20 deny 172.16.0.0 0.15.255.255
    30 deny 192.168.0.0 0.0.255.255
    40 deny 127.0.0.0 0.255.255.255
    50 permit any

ISP#
```

Corp Edge ACLs:

```
Corp-Edge(config-ext-nacl)#exit
Corp-Edge(config)#do sh ip access-lists
Standard IP access list 1
    10 permit 172.16.0.0 0.0.255.255 (866 match(es))
Extended IP access list DENY_ALLOW_CORP
    10 deny ip 10.0.0.0 0.255.255.255 any
    30 deny ip 192.168.0.0 0.0.255.255 any
    40 deny ip 127.0.0.0 0.255.255.255 any
    50 permit ip 172.16.4.0 0.0.0.3 any (40 match(es))
    60 permit ip 172.16.5.0 0.0.0.255 any (2 match(es))
    70 permit ip 172.16.6.0 0.0.0.255 any
    80 permit ip 172.16.7.0 0.0.0.255 any
    90 permit ip 172.100.10.0 0.0.0.255 any
    100 permit ip any any
    110 deny ip 172.16.0.0 0.15.255.255 any

Corp-Edge(config)#
```

ZPF-Verification

Before generating FTP and HTTP Traffic

```
Corp-Edge(config)#do show policy-map type inspect zone-pair sessions

policy exists on zp corp-to-internet
 Zone-pair: corp-to-internet

  Service-policy inspect : corp-to-internet-policy

    Class-map: corp-to-internet (match-any)
      Match: protocol ftp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol https
        0 packets, 0 bytes
        30 second rate 0 bps
      Inspect

    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
Corp-Edge(config)#
```

After generating FTP and HTTP traffic coming from the corp network going to the internet network

```
        0 packets, 0 bytes
Corp-Edge(config)#do show policy-map type inspect zone-pair sessions

policy exists on zp corp-to-internet
 Zone-pair: corp-to-internet

  Service-policy inspect : corp-to-internet-policy

    Class-map: corp-to-internet (match-any)
      Match: protocol ftp
        39 packets, 1599 bytes
        30 second rate 0 bps
      Match: protocol https
        119 packets, 4911 bytes
        30 second rate 0 bps
      Inspect

        Number of Established Sessions = 5
        Established Sessions
         Session 4068648544 (172.100.10.10:22)=>(200.16.16.10:21) tcp SIS_OPEN/TCP_ESTAB
          Created 00:00:03, Last heard  00:00:03
          Bytes sent (initiator:responder) [124:156]
         Session 4234969696 (172.16.5.5:21)=>(200.16.16.10:21) tcp SIS_OPEN/TCP_ESTAB
          Created 00:01:28, Last heard  00:01:28
          Bytes sent (initiator:responder) [124:156]
         Session 4234968160 (172.16.5.5:22)=>(200.16.16.10:21) tcp SIS_OPEN/TCP_ESTAB
          Created 00:01:24, Last heard  00:01:24
          Bytes sent (initiator:responder) [124:156]
         Session 2273238160 (172.16.6.6:21)=>(200.16.16.10:21) tcp SIS_OPEN/TCP_ESTAB
          Created 00:02:01, Last heard  00:02:01
          Bytes sent (initiator:responder) [124:156]
         Session 4288592816 (172.16.7.7:21)=>(200.16.16.10:21) tcp SIS_OPEN/TCP_ESTAB
          Created 00:03:56, Last heard  00:03:56
          Bytes sent (initiator:responder) [124:156]
    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
Corp-Edge(config)#
```

Test Cases:

Before OSPF:

Fire	Last Status	Source	Destination	Type	Color	Time(sec	Periodic	Num	Edit
	Successful	Corp-...	Corp-Mgmt-...	ICMP		0.000	N	0	(edit)
	Failed	Corp-...	Corp-Edge	ICMP		0.000	N	1	(edit)
	Successful	Corp-...	Corp-L3-SW	ICMP		0.000	N	2	(edit)
	Failed	Corp-...	Internet-HTT...	ICMP		0.000	N	3	(edit)
	Successful	ISP	Corp-Edge	ICMP		0.000	N	4	(edit)
	Failed	Corp-...	Corp-AAA-...	ICMP		0.000	N	5	(edit)

After OSPF

Fire	Last Status	Source	Destination	Type	Color	Time(sec	Periodic	Num
	Successful	Corp-HR-PC	Corp-Mgmt-...	ICMP		0.000	N	0
	Successful	Corp-Edge	Corp-HR-PC	ICMP		0.000	N	1
	Successful	Corp-HR-PC	Corp-Edge	ICMP		0.000	N	2
	Successful	Corp-HR-PC	ISP	ICMP		0.000	N	3
	Successful	Internet-PC	Corp-HR-PC	ICMP		0.000	N	4
	Successful	Internet-HTTP-Server	Corp-Mgmt-...	ICMP		0.000	N	5
	Successful	Internet-PC	Corp-Rogu...	ICMP		0.000	N	6

Before NAT and PAT

Corp-Edge

Physical Config CLI Attributes

IOS Command Line Interface

```
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 router-id 192.168.255.2
 log-adjacency-changes
 area 0 authentication message-digest
 network 172.16.4.0 0.0.0.3 area 0
 network 200.16.15.0 0.0.0.255 area 0
 network 192.168.255.2 0.0.0.0 area 0
 default-information originate
!
ip classless
ip route 0.0.0.0 0.0.0.0 200.16.15.1
!
ip flow-export version 9
!
!
!
banner motd ^CUNAUTHORIZED ACCESS PROHIBITED!!!^C
!
!
!
!
line con 0
 password 7 0822455D0A16
 logging synchronous
 login
!
line aux 0
!
line vty 0 4
 password 7 0822455D0A16
 logging synchronous
 login local
 transport input ssh
!
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec	Periodic	Num	Edit	Delete
	Successful	Internet-PC	Corp-Mgmt-PC	ICMP		0.000	N	0	(edit)	
	Successful	Corp-Rogue-PC	Corp-Dist	ICMP		0.000	N	1	(edit)	
	Successful	Corp-Rogue-PC	ISP	ICMP		0.000	N	2	(edit)	
	Successful	Internet-FTP-Server	Corp-HR-PC	ICMP		0.000	N	3	(edit)	
	Successful	Corp-AAA-Server	Internet-HTTP-Server	ICMP		0.000	N	4	(edit)	

After NAT and PAT

Corp-Edge

Physical Config CLI Attributes

IOS Command Line Interface

```
 network 172.16.4.0 0.0.0.3 area 0
 network 200.16.15.0 0.0.0.255 area 0
 network 192.168.255.2 0.0.0.0 area 0
 default-information originate
!
ip nat pool GLOBAL_POOL 200.16.16.100 200.16.16.200 netmask 255.255.255.0
ip nat inside source list 1 pool GLOBAL_POOL overload
ip classless
ip route 0.0.0.0 0.0.0.0 200.16.15.1
!
ip flow-export version 9
!
!
access-list 1 permit 172.16.0.0 0.0.255.255
!
banner motd ^CUNAUTHORIZED ACCESS PROHIBITED!!!^C
!
!
!
!
line con 0
 password 7 0822455D0A16
 logging synchronous
 login
!
line aux 0
!
line vty 0 4
 password 7 0822455D0A16
 logging synchronous
```

```
Corp-Dist# sh ip nat translation
Corp-Dist# sh ip nat translations
Corp-Dist#
Corp-Dist#
Corp-Dist#
Corp-Dist#
Corp-Dist# sh ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 200.16.16.100:3   172.16.4.6:3       172.16.6.6:3       172.16.6.6:3
icmp 200.16.16.100:4   172.16.4.6:4       200.16.16.10:4     200.16.16.10:4
icmp 200.16.16.100:4   172.16.4.6:4       200.16.16.5:4      200.16.16.5:4

Corp-Dist#
```

Before ACL configurations

Fire	Last Status	Source	Destination	Type	Color	Time(sec	Periodic	Num
	Successful	Corp-HR-PC	Corp-Mgmt-...	ICMP		0.000	N	0
	Successful	Corp-Edge	Corp-HR-PC	ICMP		0.000	N	1
	Successful	Corp-HR-PC	Corp-Edge	ICMP		0.000	N	2
	Successful	Corp-HR-PC	ISP	ICMP		0.000	N	3
	Successful	Internet-PC	Corp-HR-PC	ICMP		0.000	N	4
	Successful	Internet-HTTP-Server	Corp-Mgmt-...	ICMP		0.000	N	5
	Successful	Internet-PC	Corp-Rogu...	ICMP		0.000	N	6

After ACLs

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec	Periodic	Num	Edit	De
	Successful	Corp-HR-PC	Internet-PC	ICMP		0.000	N	0	(edit)	
	Failed	Internet-PC	Corp-HR-PC	ICMP		0.000	N	1	(edit)	
	Successful	Corp-HR-PC	Internet-PC	ICMP		0.000	N	2	(edit)	
	Failed	Internet-PC	Corp-Rogue-PC	ICMP		0.000	N	3	(edit)	
	Successful	Corp-HR-PC	Internet-FTP-Server	ICMP		0.000	N	4	(edit)	
	Successful	Internet-HTTP-S...	Corp-Dist	ICMP		0.000	N	5	(edit)	
	Successful	Corp-Rogue-PC	ISP	ICMP		0.000	N	6	(edit)	

Before ZPF configurations:

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec	Periodic	Num	Edit	De
	Successful	Corp-HR-PC	Internet-PC	ICMP		0.000	N	0	(edit)	
	Failed	Internet-PC	Corp-HR-PC	ICMP		0.000	N	1	(edit)	
	Successful	Corp-HR-PC	Internet-PC	ICMP		0.000	N	2	(edit)	
	Failed	Internet-PC	Corp-Rogue-PC	ICMP		0.000	N	3	(edit)	
	Successful	Corp-HR-PC	Internet-FTP-Server	ICMP		0.000	N	4	(edit)	
	Successful	Internet-HTTP-S...	Corp-Dist	ICMP		0.000	N	5	(edit)	
	Successful	Corp-Rogue-PC	ISP	ICMP		0.000	N	6	(edit)	

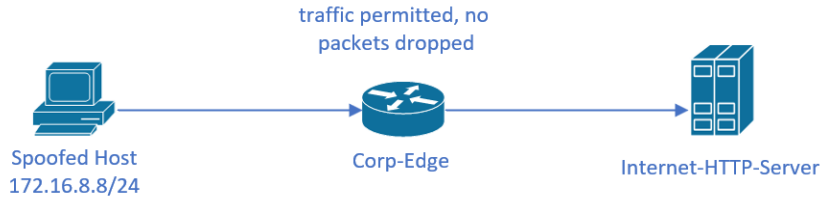
After ZPF Configurations:

Fire	Last Status	Source	Destination	Type
	Successful	Corp-HR-PC	Corp-Rogue-PC	ICMP
	Failed	Corp-HR-PC	ISP	ICMP
	Failed	Corp-HR-PC	Internet-PC	ICMP
	Failed	Internet-PC	Corp-Mgmt-PC	ICMP
	Successful	Corp-HR-PC	Corp-Edge	ICMP
	Successful	Internet-FTP-Server	Corp-Edge	ICMP
	Failed	Internet-FTP-Server	Corp-HR-PC	ICMP
	Failed	Internet-FTP-Server	Corp-Dist	ICMP
	Failed	Corp-HR-PC	Internet-PC	ICMP
	Failed	Internet-HTTP-Server	Corp-Rogue-PC	ICMP
	Failed	Corp-HR-PC	Internet-FTP-Server	ICMP

Test Cases

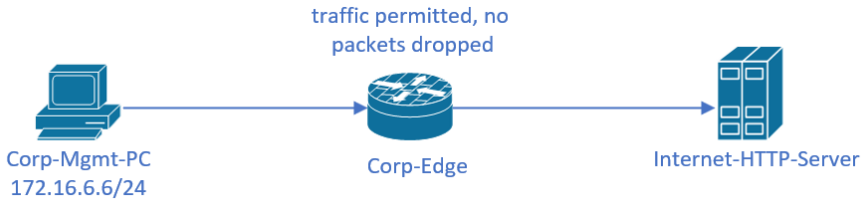
01. Before Configuring Spoofing Mitigation Techniques

a. Any host with an IP address seeming to originate from the Corporate LAN can reach the internet.

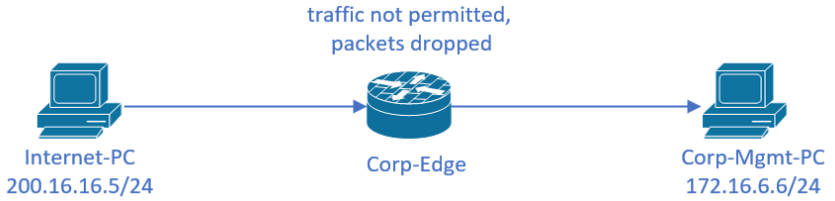


02. After Configuring Spoofing Mitigation Techniques

a. Only hosts within the Corporate LAN should be able to reach the Corp-Edge router.

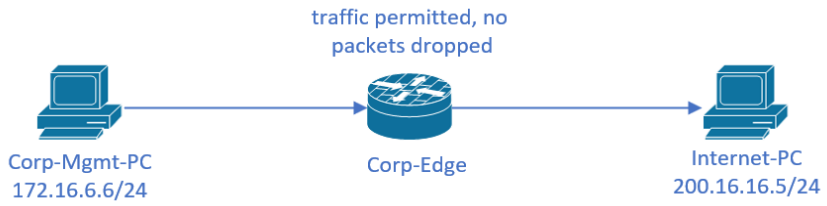


b. Hosts on the internet should not be able to reach hosts within the Corporate LAN.



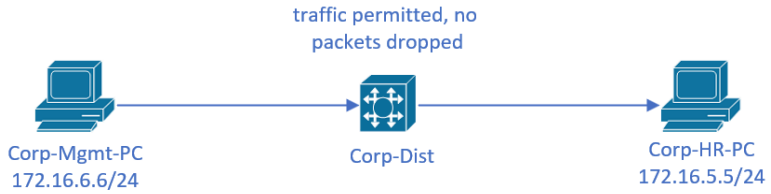
03. Before Configuring a Zone-Based Policy Firewall

a. Hosts on the Corporate LAN should be able to reach hosts on the internet.



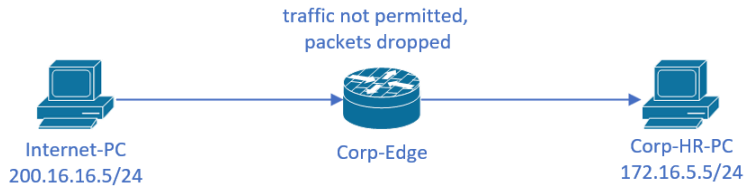
04. After Configuring a Zone-Based Policy Firewall

a. All hosts within the Corporate LAN should be able to reach each other.



b. All hosts on the Internet should be able to reach the Corp-Edge.

c. Hosts on the internet and hosts within the Corporation LAN should not be able to reach each other.



Conclusion:

This lab focuses on the implementation of zone-based firewall technology, access control lists, and NAT/PAT to try to protect networks from external threats coming from the internet. Limiting what protocols can be used, mitigating spoofing attacks, and how packet filters can be used to mitigate attacks are all explored in this lab. With the implementation of these different tools, we were able to mitigate the different kinds of attacks demonstrated in this lab.